Passive Construction Specification

ClimateAdapt-Comfort Decision-Support System | Builder-Ready Specification

Simulated City: Mumbai	Building Typology: Shelter_v1
Confidence Tier: Low	Verification Status: Verified
Location Note: Matched to simulated climate station: Mumbai (0.0 km).	Package: Balanced Package

1. Recommended Envelope Design Variables

Parameter	Specification	Engineering Units	Physical Rationale
Roof Insulation	150	mm	Attenuates extreme solar heat gain through slab
Wall Insulation	42	mm	Buffers exterior ambient convective heat flux
Roof Solar Absorptance	0.60	dimensionless (0-1)	Cool-roof coating reflects solar radiation
Window-to-Wall Ratio	16.5	%	Balanced daylighting vs conductive heat gain
Orientation	356	degrees (° from N)	Optimizes seasonal solar orientation
Overhang Shading Depth	0.98	meters (m)	Blocks high summer sun while allowing winter light
Natural Cooling Ventilation	1.0	ACH (Air Changes/hr)	Night-time flush removes stored daytime mass heat
Wall Construction Mass	Medium	category	Thermal mass shifts diurnal peak cooling loads

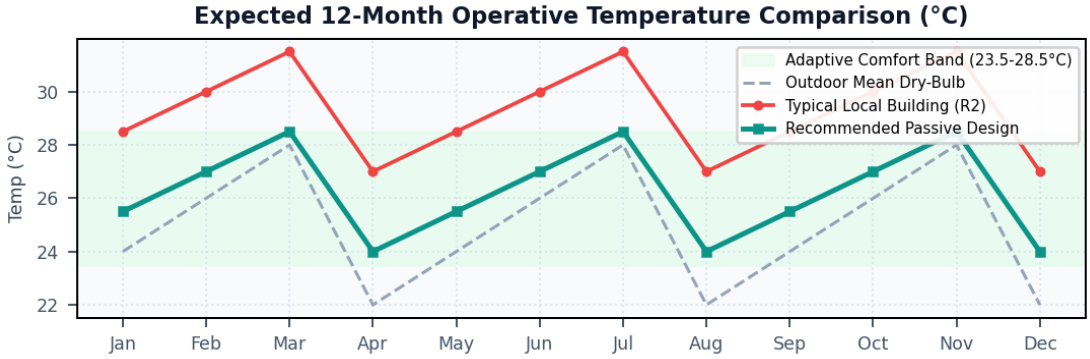
2. Bill of Materials (BOM) Breakdown

Item ID	Description	Quantity	Unit	Category
-	Common burnt clay brick masonry 1:6	45.1	m2	wall_masonry
-	Expanded polystyrene (EPS) / minera	1.9	m3	insulation
-	Reinforced cement concrete 150mm ro	20.2	m2	roof_structure
-	Expanded polystyrene (EPS) board ro	3.0	m3	insulation
-	Standard cementitious water-proofin	20.2	m2	roof_finish
-	Powder coated aluminium window fram	8.9	m2	glazing
-	Cantilever horizontal exterior wind	7.3	m2	shading
-	Operable glazed ventilation louvers	0.9	m2	ventilation

3. Rough Cost Range Estimate (CPWD DSR 2021)

Official Pricing Statement: Estimated Budget Range: Unavailable (Rates unpopulated in official DSR schedule).
Cost numbers are withheld to prevent price fabrication until officially verified.
Schedule Basis: CPWD DSR 2021. Estimate for early decisions, never a quote. Actual contractor tenders depend on local market conditions, transport, site logistics, and GST.

4. Expected Thermal & Energy Performance



Annual Discomfort (ML Champion): 0.0% hours | Physical Adaptive 90%: 0.0% hours | Dual-Judge Disagreement: 0.0 pts | HVAC Energy (O2): 0.0 kWh/m²/yr

5. Confidence Tier & Research Limitations

1. Simulation-based estimate using EnergyPlus 26.2.0; does not substitute for licensed MEP or structural design.
2. Cost ranges represent early envelope comparisons using CPWD DSR rate models and do not constitute a commercial quotation.
3. Occupant comfort boundaries are derived from ASHRAE DB II empirical survey data and ASHRAE 55 Adaptive 90% envelopes.
4. Future climate projections reflect temperature shifts and psychrometric morphing, not regional dynamic downscaling.